STAT GU4204 Section 001 — Mock Midterm Exam

Spring 2026 Total: 100 points Time: 55 minutes

Name: _____ UNI: _____

Instructions: Show work to get full credit. You do NOT need to simplify your answers.

You can use, throughout the exam, any of the following values if needed:

$$z_{0.025} = 1.96 \qquad z_{0.05} = 1.645 \qquad z_{0.005} = 2.576 \qquad t_{0.025,\,24} = 2.064 \qquad t_{0.025,\,15} = 2.131$$

Part A: Fill in the Blank (10 points, 2 points each)

Question 1 (2 points) Fill in the Blank

A function $T = \varphi(X_1, \ldots, X_n)$ that depends only on the observed data and contains no unknown parameters is called a _____.

Question 2 (2 points) Fill in the Blank

The decomposition $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [\mathrm{bias}(\hat{\theta})]^2$ is called the _____ decomposition.

Question 3 (2 points) Fill in the Blank

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} a$ (a constant), then $X_n Y_n \xrightarrow{d} aX$. This result is known as _____ theorem.

Question 4 (2 points) Fill in the Blank

A prior distribution such that the posterior distribution belongs to the same parametric family is called a _____ prior.

Question 5 (2 points) Fill in the Blank

An unbiased estimator whose variance equals the Cramér-Rao lower bound is called an _____ estimator.

Part B: True or False (12 points, 2 points each)

Circle T or F. No justification required.

Question 6 (2 points) **True/False**

T **F** If $\hat{\theta}_n$ is a consistent estimator of θ, then $\hat{\theta}_n$ is unbiased for θ for every n.

Question 7 (2 points) **True/False**

T **F** Any one-to-one function of a sufficient statistic is also a sufficient statistic.

Question 8 (2 points) **True/False**

T **F** If (3.1, 5.7) is a 95% confidence interval for μ, then $P(\mu \in (3.1, 5.7)) = 0.95$.

Question 9 (2 points) **True/False**

T **F** For $X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$, the Central Limit Theorem guarantees that $\bar{X}_n$ has an exact normal distribution.

Question 10 (2 points) **True/False**

T **F** The Cramér-Rao lower bound can be applied when $X_1, \ldots, X_n$ are iid Uniform$[0, \theta]$.

Question 11 (2 points) **True/False**

T **F** For $X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$, the sample mean $\bar{X}$ and the sample variance s^2 are independent random variables.

Part C: Multiple Choice (9 points, 3 points each)

Question 12 (3 points) **Multiple Choice**

Let $X_1, \ldots, X_n$ be iid Poisson(θ). The maximum likelihood estimator of θ is:

(a) $\frac{1}{n}\sum_{i=1}^{n} X_i^2$

(b) $\frac{1}{\bar{X}_n}$

(c) $\bar{X}_n$

(d) $\dfrac{n-1}{n}\bar{X}_n$

Question 13 (3 points) **Multiple Choice**

Let $X_1, \ldots, X_n$ be iid from a distribution with mean μ and finite variance σ^2. The Weak Law of Large Numbers requires which conditions?

(a) iid and finite mean only

(b) iid and finite variance only

(c) iid, finite mean, and normality of X_i

(d) iid and finite mean (finite variance is sufficient but not necessary for the WLLN)

Question 14 (3 points) **Multiple Choice**

Which of the following is a sufficient statistic for θ when $X_1, \ldots, X_n$ are iid Exponential(θ) with pdf $f(x;\theta) = \theta e^{-\theta x}$ for $x > 0$?

(a) $X_{(n)} = \max(X_1, \ldots, X_n)$

(b) $X_{(1)} = \min(X_1, \ldots, X_n)$

(c) $\displaystyle\sum_{i=1}^{n} X_i$

(d) $\displaystyle\prod_{i=1}^{n} X_i$

Part D: Short Answer (9 points)

Question 15 (3 points) **Listing**

State the three regularity conditions (A.1, A.2, A.3) required for the Cramér-Rao lower bound to hold.

Question 16 (3 points) **Short Answer**

Explain, in one or two sentences, why a confidence interval should be interpreted using the word "confidence" and not "probability." What is the conceptual error in saying "θ lies in the interval with probability 0.95"?

Question 17 (3 points) **Short Answer**

State the Central Limit Theorem precisely. Be sure to include all conditions.

Part E: Computational Problems (60 points)

Question 18 (15 points) **MLE, Bias, and Sufficiency**

Let $X_1, X_2, \ldots, X_n$ be a random sample from a Poisson distribution with probability mass function

$$f(x;\theta) = \frac{e^{-\theta}\,\theta^x}{x!}, \quad x = 0, 1, 2, \ldots$$

(a) (4 points) Find the maximum likelihood estimator $\hat{\theta}$ of θ.

(b) (3 points) Is $\hat{\theta}$ unbiased for θ? Justify your answer.

(c) (4 points) Using the factorization criterion, show that $T = \sum_{i=1}^{n} X_i$ is a sufficient statistic for θ.

(d) (4 points) Find the Fisher information $I_n(\theta)$ in the sample. Is $\hat{\theta}$ an efficient estimator of θ?

Question 19 (12 points) **Delta Method**

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean $\theta > 0$ and variance σ^2.

(a) (4 points) State the asymptotic distribution of $\bar{X}_n$ (i.e., its approximate distribution for large n).

(b) (8 points) Using the delta method, determine the asymptotic distribution of $g(\bar{X}_n) = 1/\bar{X}_n$ for large n. Clearly state the mean and variance of the asymptotic distribution.

Question 20 (12 points) **Bayesian Inference**

Let $X_1, X_2, \ldots, X_n$ be a random sample from a Bernoulli(θ) distribution, where $\theta \in [0, 1]$. Suppose the prior distribution on θ is Beta(α, β), with density

$$\xi(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\,\theta^{\alpha-1}(1-\theta)^{\beta-1}, \quad 0 < \theta < 1.$$

(a) (5 points) Show that the posterior distribution of θ given the data is Beta($\alpha + \sum x_i,\ \beta + n - \sum x_i$).

(b) (3 points) What is the Bayes estimator of θ under squared error loss?

(c) (4 points) Show that as $n \to \infty$, the Bayes estimator converges to the MLE.

Question 21 (12 points) **Confidence Intervals and Sampling Distributions**

Let $X_1, X_2, \ldots, X_{25}$ be a random sample from a $N(\mu, \sigma^2)$ distribution. Suppose $\bar{x} = 12.4$ and $s^2 = 9$.

(a) (3 points) Construct an exact 95% confidence interval for μ.

(b) (3 points) If σ^2 were known to equal 9, construct a 95% confidence interval for μ and compare its width to the interval in part (a).

(c) (3 points) How large a sample size would be needed (with $\sigma^2 = 9$ known) so that the 95% confidence interval has a margin of error no greater than 0.5?

(d) (3 points) State the distribution of $\dfrac{\sum_{i=1}^{25}(X_i - \bar{X})^2}{\sigma^2}$ and use it to give a 95% confidence interval for σ^2.

Question 22 (9 points) **Probability Inequalities and Convergence**

Let $X_1, X_2, \ldots, X_n$ be iid with mean μ and variance σ^2.

(a) (3 points) Using Chebyshev's inequality, find an upper bound on $P(|\bar{X}_n - \mu| \geq 2\sigma)$.

(b) (3 points) How large must n be so that $P(|\bar{X}_n - \mu| \leq 0.5\sigma) \geq 0.99$?

(c) (3 points) Using the Weak Law of Large Numbers and the Continuous Mapping Theorem, show that $\bar{X}_n^2 \xrightarrow{P} \mu^2$.

— End of Mock Midterm Exam —

Answer Key

Part A: Fill in the Blank

1. **statistic**
2. **bias-variance**
3. **Slutsky's**
4. **conjugate**
5. **efficient**

Part B: True or False

6. **F** — Consistency is a large-n property ($\hat{\theta}_n \xrightarrow{P} \theta$). An estimator can be biased at every finite n and still be consistent. Example: $X_{(n)}$ for Unif$[0, \theta]$ is biased but consistent.
7. **T** — If T is sufficient for θ and h is one-to-one, then $h(T)$ is also sufficient. This follows from the factorization criterion: if $f(\mathbf{x}, \theta) = u(\mathbf{x})\, v(T(\mathbf{x}), \theta)$, then we can write the same factorization in terms of $h(T)$ since h is invertible.
8. **F** — θ is a fixed (unknown) constant, not a random variable. After observing the data, the interval either contains θ or it doesn't. The correct statement uses "confidence," not "probability."
9. **F** — For normal data, $\bar{X}_n$ is *exactly* normal by the reproductive property of the normal distribution (Corollary 5.6.2), not by the CLT. The CLT provides only an approximation and is not needed here.
10. **F** — Regularity condition A.1 requires the support $\{x : f(x, \theta) > 0\}$ to not depend on θ. For Unif$[0, \theta]$, the support is $[0, \theta]$, which changes with θ. Hence CRLB does not apply.
11. **T** — This is Proposition 6.3. Moreover, this independence *characterizes* the normal distribution: F is normal if and only if $\bar{X}$ and s^2 are independent for all n.

Part C: Multiple Choice

12. **(c)** $\bar{X}_n$. The log-likelihood is $\ell(\theta) = -n\theta + (\sum x_i) \log \theta - \sum \log(x_i!)$. Setting $\ell'(\theta) = -n + \sum x_i/\theta = 0$ gives $\hat{\theta} = \bar{X}_n$.
13. **(d)** The WLLN requires iid and $E|X_1| < \infty$ (finite mean). Finite variance is sufficient to prove WLLN via Chebyshev, but is not necessary for the result to hold.
14. **(c)** $\sum X_i$. Joint pdf: $\prod \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum x_i}$. This factors as $\underbrace{1}_{u(\mathbf{x})} \cdot \underbrace{\theta^n e^{-\theta \sum x_i}}_{v(\sum x_i, \theta)}$, so $\sum X_i$ is sufficient by the factorization criterion.

Part D: Short Answer

15. **Regularity Conditions**:

    (A.1) The support $\{x : f(x,\theta) > 0\}$ does not depend on θ.

    (A.2) Differentiation and integration can be interchanged: $\frac{\partial}{\partial\theta}\int W(\mathbf{x})\, f(\mathbf{x},\theta)\, d\mathbf{x} = \int W(\mathbf{x})\, \frac{\partial f}{\partial\theta}(\mathbf{x},\theta)\, d\mathbf{x}$.

    (A.3) $\frac{\partial}{\partial\theta}\log f(x,\theta)$ exists for all x in the support and all $\theta \in \Omega$, as a finite quantity.

16. **CI Interpretation**: θ is a fixed unknown constant, not a random variable. The interval (A, B) is random (it depends on the data). Before observing data, we can say that the random interval will capture θ with probability $1-\alpha$. But after observing specific values $(3.1, 5.7)$, the interval either contains θ or it doesn't—there is no more randomness. Hence we say "confidence," not "probability."

17. **CLT**: Let $X_1, X_2, \ldots$ be iid random variables with $E(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty.$$

    Conditions: (1) iid, (2) finite variance.

Part E: Computational Problems

Question 18 (Poisson MLE, Bias, Sufficiency, Fisher Info)

(a) **MLE**: $L(\theta) = \prod_{i=1}^n \frac{e^{-\theta}\theta^{x_i}}{x_i!} = \frac{e^{-n\theta}\,\theta^{\sum x_i}}{\prod x_i!}$.

$\ell(\theta) = -n\theta + (\sum x_i)\log\theta - \sum \log(x_i!)$.

$\ell'(\theta) = -n + \frac{\sum x_i}{\theta} = 0 \implies \boxed{\hat{\theta} = \bar{X}_n}$.

$\ell''(\theta) = -\sum x_i/\theta^2 < 0$, confirming maximum.

(b) **Unbiasedness**: $E(\hat{\theta}) = E(\bar{X}_n) = E(X_1) = \theta$ for all θ. So $\hat{\theta} = \bar{X}_n$ is **unbiased**.

(c) **Sufficiency**: The joint pmf is

$$f(\mathbf{x};\theta) = \frac{e^{-n\theta}\,\theta^{\sum x_i}}{\prod x_i!} = \underbrace{\frac{1}{\prod x_i!}}_{u(\mathbf{x})} \cdot \underbrace{e^{-n\theta}\,\theta^{\sum x_i}}_{v(\sum x_i,\,\theta)}$$

By the factorization criterion, $T = \sum_{i=1}^n X_i$ is sufficient for θ.

(d) **Fisher Information**: For one observation:

$$\log f(x;\theta) = -\theta + x\log\theta - \log(x!)$$

$$\dot{\ell}(x,\theta) = \frac{\partial}{\partial\theta}\log f = -1 + \frac{x}{\theta} = \frac{x-\theta}{\theta}$$

$$I_1(\theta) = E\left[\left(\frac{X-\theta}{\theta}\right)^2\right] = \frac{\mathrm{Var}(X)}{\theta^2} = \frac{\theta}{\theta^2} = \frac{1}{\theta}$$

$$I_n(\theta) = \frac{n}{\theta}$$

CRLB for θ: $\text{Var}(T) \geq 1/I_n(\theta) = \theta/n$.

$\text{Var}(\hat{\theta}) = \text{Var}(\bar{X}_n) = \theta/n = \text{CRLB}$.

Since $\hat{\theta}$ is unbiased and achieves the CRLB, it is **efficient** (and therefore MVUE).

Question 19 (Delta Method)

(a) By the CLT, for large n:

$$\bar{X}_n \sim_{\text{approx}} N\left(\theta,\ \frac{\sigma^2}{n}\right)$$

Equivalently: $\frac{\sqrt{n}(\bar{X}_n - \theta)}{\sigma} \xrightarrow{d} N(0,1)$.

(b) Let $g(x) = 1/x$. Then $g'(x) = -1/x^2$, so $g'(\theta) = -1/\theta^2 \neq 0$ (since $\theta > 0$).

By the delta method:

$$\sqrt{n}\left(\frac{1}{\bar{X}_n} - \frac{1}{\theta}\right) \xrightarrow{d} N\big(0,\ \sigma^2 \cdot [g'(\theta)]^2\big) = N\left(0,\ \frac{\sigma^2}{\theta^4}\right)$$

Therefore, the asymptotic distribution of $1/\bar{X}_n$ is:

$$\boxed{\frac{1}{\bar{X}_n} \sim_{\text{approx}} N\left(\frac{1}{\theta},\ \frac{\sigma^2}{n\theta^4}\right)}$$

Question 20 (Bayesian Inference)

(a) The likelihood is:

$$f(\mathbf{x}|\theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}$$

Posterior $\propto$ Likelihood $\times$ Prior:

$$\xi(\theta|\mathbf{x}) \propto \theta^{\sum x_i}(1-\theta)^{n-\sum x_i} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1} = \theta^{\alpha+\sum x_i - 1}(1-\theta)^{\beta+n-\sum x_i - 1}$$

This is the kernel of a $\boxed{\text{Beta}(\alpha + \sum x_i,\ \beta + n - \sum x_i)}$ distribution.

(b) Under squared error loss, the Bayes estimator is the **posterior mean**:

$$\hat{\theta}_{\text{Bayes}} = E[\theta|\mathbf{X}] = \frac{\alpha + \sum X_i}{\alpha + \beta + n}$$

(c) As $n \to \infty$:

$$\hat{\theta}_{\text{Bayes}} = \frac{\alpha + \sum X_i}{\alpha + \beta + n} = \frac{\alpha/n + \bar{X}_n}{(\alpha+\beta)/n + 1} \xrightarrow{n\to\infty} \frac{0 + \bar{X}_n}{0+1} = \bar{X}_n = \hat{\theta}_{\text{MLE}}$$

The prior parameters α, β become negligible relative to the sample size, so the Bayes estimator converges to the MLE. (Data overwhelms the prior.)

Question 21 (Confidence Intervals)

(a) σ^2 unknown $\Rightarrow$ use t-distribution. Pivot: $\frac{\sqrt{n}(\bar{X}-\mu)}{s} \sim t_{n-1} = t_{24}$.
95% CI: $\bar{x} \pm t_{0.025,\,24} \cdot \frac{s}{\sqrt{n}} = 12.4 \pm 2.064 \cdot \frac{3}{\sqrt{25}} = 12.4 \pm 2.064 \cdot 0.6 = 12.4 \pm 1.238$.

$$\boxed{(11.162,\ 13.638)}$$

(b) $\sigma^2 = 9$ known $\Rightarrow$ use z-distribution.
95% CI: $\bar{x} \pm z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} = 12.4 \pm 1.96 \cdot \frac{3}{5} = 12.4 \pm 1.176$.

$$\boxed{(11.224,\ 13.576)}$$

This interval is **narrower** than part (a) because $z_{0.025} = 1.96 < t_{0.025,24} = 2.064$. The t-interval is wider to compensate for the additional uncertainty from estimating σ.

(c) Margin of error: $z_{0.025} \cdot \sigma/\sqrt{n} \leq 0.5$.
$1.96 \cdot 3/\sqrt{n} \leq 0.5 \;\Rightarrow\; \sqrt{n} \geq 1.96 \cdot 6 = 11.76 \;\Rightarrow\; n \geq 138.3$.

$$\boxed{n \geq 139}$$

(d) $\frac{\sum(X_i-\bar{X})^2}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} = \chi^2_{24}$.
95% CI for σ^2: $\left(\frac{(n-1)s^2}{\chi^2_{24,\,0.025}},\ \frac{(n-1)s^2}{\chi^2_{24,\,0.975}}\right)$.
Using $\chi^2_{24,\,0.025} = 39.36$ and $\chi^2_{24,\,0.975} = 12.40$:

$$\boxed{\left(\frac{24 \cdot 9}{39.36},\ \frac{24 \cdot 9}{12.40}\right) = (5.49,\ 17.42)}$$

(Note: in an actual exam, the χ^2 table values would be provided.)

Question 22 (Probability Inequalities and Convergence)

(a) $P(|\bar{X}_n - \mu| \geq 2\sigma) \leq \frac{\mathrm{Var}(\bar{X}_n)}{(2\sigma)^2} = \frac{\sigma^2/n}{4\sigma^2} = \boxed{\frac{1}{4n}}$

(b) Need $P(|\bar{X}_n - \mu| \geq 0.5\sigma) \leq 0.01$.
By Chebyshev: $\frac{\sigma^2/n}{(0.5\sigma)^2} = \frac{4}{n} \leq 0.01 \;\Rightarrow\; n \geq 400$.

$$\boxed{n \geq 400}$$

(c) By WLLN: $\bar{X}_n \xrightarrow{P} \mu$ (requires iid, finite mean).
Let $g(x) = x^2$, which is continuous everywhere (in particular, at $x = \mu$).
By the Continuous Mapping Theorem: $g(\bar{X}_n) = \bar{X}_n^2 \xrightarrow{P} g(\mu) = \mu^2$. $\square$